Calc III Sections

Fall 2025

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Calc III-Week 10 (10/27-31)

vector fields, divergence and curl, double, triple integral Topics: (1) Acceleration and Arc Length, (2) Vector Fields. computing double and triple integrals

Definition 0.1 (flow line). Let F be a vector field, a flow line of F is a path $c(t)$ satisfying

$$c'(t) = F(c(t))$$

(Tangent vector of the path coincides with the given vector field F).

Definition 0.2 (divergence). Let F be a vector field in $\mathbb{R}^3$ $F = (F_1, F_2, F_3)$, the divergence of F is the **scalar field** (assigns one number to an given point (x, y, z)),

$$\operatorname{div} F := \nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

More generally, if $F = (F_1, \dots, F_n)$ is a vector field on $\mathbb{R}^n$, its divergence is

$$\operatorname{div} F = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \cdots + \frac{\partial F_n}{\partial x_n}$$

Remark 1. *We write the divergence as $\nabla \cdot F$ because*

$$\nabla = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right)$$

and if $F = (F_1, \dots, F_n)$,

$$\operatorname{div} F = \frac{\partial F_1}{\partial x_1} + \cdots + \frac{\partial F_n}{\partial x_n} = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right) \cdot (F_1, \dots, F_n) = \nabla \cdot F$$

Definition 0.3 (curl). Let F be a vector field in $\mathbb{R}^3$, writing $F = (F_1, F_2, F_3)$, the **curl** of F is the vector field

$$\operatorname{curl} F := \nabla \times F = \det \begin{bmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix}$$

If $\operatorname{curl} F = 0$, then we say the vector field is **irrotational**.

Proposition 0.1 (gradient is irrotational). Let $f \in C^2$, viewing ∇f as a vector field, then

$$\nabla \times (\nabla f) = 0$$

Proposition 0.2 (divergence of a curl vanishes). For any C^2 vector field F,

$$\nabla \cdot (\nabla \times F) = 0$$

Problem 0.1. Show that the vector field $V(x, y, z) = (x^2, -y, z)$ is not the curl of any vector field F. In other words, there is no vector field F such that

$$V = \operatorname{curl} F$$

Problem 0.2 (Marsden-Tromba, IV.4, 21)**.** Let $F(x, y, z) = (x^2, x^2y, z + zx)$.

(a) Verify that $\nabla \cdot (\nabla \times F) = 0$.

(b) Can there exist a function $f : \mathbb{R}^3 \to \mathbb{R}$ such that $F = \nabla f$.